NATIONAL INSTITUTE OF TECHNOLOGY, SRINAGAR

COMPUTER ORGANIZATION AND ARCHITECTURE

Assignment 1

Submission policies (Due Date 18th August)

- A maximum of two students are allowed to form a group for the assignment.
- A document created using latex is to be submitted. The document should contain all the required solutions with neat and clean diagrams.
- Plagiarism is strictly prohibited. If any plagiarism is found the students will be given zero marks in all the future assignments.

Files in this assignment:

- ***add.c* and *add.s***
- ***factorial.c* and *factorial.s***

1. Learn to use ARM cross compiler. Write in your own words about cross compilation.
2. Install Ubuntu on your PC. Install arm-linux-gnueabi cross compiler.
3. Generate the assembly code for the hello-world.c program using the cross-compiler.
4. Generate the executable of the program using the cross-compiler.
5. Learn about the qemu tool and install on your system.
6. Run the generated executable using qemu tool.
7. For each line of the program, add.s, explain each line of assembly code.

   Example:

   mov r3, #4 Set the value of register, r3, to the constant, 4.

8. Trace the execution of the program factorial.s.

All the relevant explanation about your answers and the softwares should be provided in the final document. Even the assembly code of helloworld.c should be provided in the document.

Program Tracing

Consider an assembly (.s) file. Start with the first line of the main function. For each assembly instruction, you need to write the values of the source registers, memory values, and destination registers in a tabular format. This table shows some sample entries for the add.s file. Please note that for stmfd and ldmfd instructions, you need to break them into individual loads and stores.

Instruction	dest.		src1.		src2.	
	location	value	location	value	location	value
str r3, [fp, #-16]	address	84	r3	4	fp	100
ldr r3, [fp, #-20]	r3	6	address	80	fp	100
add r3, r2, r3	r3	10	r2	4	r3	6
ldr r0, .L2	r0	.LC0	address	.L2		
bl printf	lr	line 33	function	printf		